

**Big Idea.** By the end of Chapter 11, we will be able to approximate *any function* by a polynomial, called its *Taylor polynomial*. Polynomials are easy to integrate, so this will help us to evaluate difficult/impossible integrals.

**Definition.** We say a sequence  $\{a_n\}$  **converges** to  $L$  if  $a_n \rightarrow L$  as  $n \rightarrow \infty$ , that is, if  $\lim_{n \rightarrow \infty} a_n = L$ . If the limit  $\lim_{n \rightarrow \infty} a_n$  does not exist, we say  $\{a_n\}$  **diverges**.

**Theorem.** If  $\lim_{x \rightarrow \infty} f(x)$  exists, then the sequence  $a_n = f(n)$  converges to the same limit:

$$\lim_{n \rightarrow \infty} a_n = \lim_{x \rightarrow \infty} f(x)$$

1. Let  $a_n = r^n$  for some real number  $r$  (so  $\{a_n\}$  is a **geometric sequence**). What is the limit of the sequence  $\{a_n\}$  for the following values of  $r$ ? Can you say in general for which values of  $r$  the sequence converges?

(a)  $r = 5$

(b)  $r = 1$

(c)  $r = \frac{1}{2}$

(d)  $r = 0$

(e)  $r = -\frac{1}{3}$

(f)  $r = -4$

The following theorem says we may “bring a limit of a sequence inside a continuous function.”

**Theorem.** If  $f$  is continuous and  $\lim_{n \rightarrow \infty} a_n = L$ , then

$$\lim_{n \rightarrow \infty} f(a_n) = f\left(\lim_{n \rightarrow \infty} a_n\right) = f(L)$$

2. Use this theorem to find the limit of the following sequences: Start with  $n = 1$

(a)  $a_n = \cos\left(\frac{\pi n^4}{3 + n^4}\right)$

(b)  $b_n = \sqrt{5 + \frac{2}{n}}$

(c)  $c_n = e^{\frac{4n}{n^2+1}}$

**Definition.** A sequence  $\{a_n\}$  is *bounded above* by  $M$  if  $a_n \leq M$  for all  $n$ .  $\{a_n\}$  is *bounded below* by  $m$  if  $a_n \geq m$  for all  $n$ .  $\{a_n\}$  is *bounded* if it is both bounded below and bounded above.

3. Classify the sequences in Problem 1 as bounded above, bounded below, bounded, or none of these.

**Definition.** A sequence  $\{a_n\}$  is *increasing* if  $a_n \leq a_{n+1}$  for all  $n$ .  $\{a_n\}$  is *decreasing* if  $a_n \geq a_{n+1}$  for all  $n$ .  $\{a_n\}$  is *monotonic* if it is either increasing or decreasing.

4. Classify the sequences in Problem 1 as increasing, decreasing, or neither.

5. If a sequence  $\{a_n\}$  converges, does it have to be bounded?

6. If a sequence  $\{a_n\}$  is bounded, does it have to converge?

7. If a sequence  $\{a_n\}$  is bounded and monotonic, does it have to converge?